# Case Study: Cycling Traffic Data in Flanders

*Group 4*

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# 1 Introduction

## 1.1 Background and Motivation

Cycling is a cornerstone of sustainable mobility in Flanders, yet it remains disproportionately dangerous. In Flanders, cyclists accounted for 30% of all road fatalities in 2024 — the second largest group after car occupants — while fatalities among all other road user types have declined significantly since 2005, cyclist deaths have increased by 15% over the same period [Statistiek Vlaanderen, 2024].

To monitor cycling infrastructure, the Flemish road agency Agentschap Wegen en Verkeer (AWV) operates 151 automatic bicycle counting stations across Flanders (*fietstellingen*), recording cyclist counts at 15-minute intervals. AWV also maintains a *Gevaarlijke Punten* (dangerous points) list identifying locations with a history of serious accidents [Agentschap Wegen en Verkeer, 2025]. However, this list is inherently reactive: a location must first accumulate recorded accidents before being flagged, and it does not account for cycling exposure — a busy junction with ten accidents may be safer per cyclist than a quiet path with two.

## 1.2 Problem Statement

This project addresses that gap by developing a predictive cycling risk assessment model for AWV sensor locations. The model uses lagged traffic volume, weather conditions, and accident history from the preceding month to forecast whether a location will experience elevated risk in the following month. Because all features are drawn from the period prior to the prediction window, the model is genuinely forward-looking and avoids data leakage from concurrent observations. The analysis is scoped to the 151 AWV EcoCounter sensor locations, which provide the exposure data necessary to compute normalised risk estimates.

## 1.3 Research Objectives

1. Produce reliable next-month crash risk predictions for AWV sensor locations across Flanders.
2. Generate historical and forward-looking safety rankings, identifying sites where accident risk is disproportionate to cycling exposure.
3. Develop an interactive web dashboard that visualises historical and forecast safety rankings across AWV sensor locations, and provides a station-level prediction engine allowing users to adjust traffic and weather inputs to simulate next-month crash risk.

# 2 Data Sources and Preprocessing

## 2.1 Data Sources

### 2.1.1 Cycling Traffic Data (Fietstellingen)

The *Fietstellingen* dataset comprises monthly count records (`data-YYYY-MM.csv`) containing 15-minute interval cyclist counts per sensor, a site metadata file (`sites.csv`) providing the geographic coordinates, municipality, and installation date of each EcoCounter station, and a directional configuration file (`richtingen.csv`).

### 2.1.2 Historical Accident Data

The *Geolocation of Road Traffic Accidents 2017–2024* dataset published by Statbel [Statbel, 2025] records all police-reported road traffic accidents in Belgium with geographic coordinates. Each record includes the year and month of collision, road user types, weather and road conditions at the time of the accident, and the municipality of occurrence.

### 2.1.3 Weather Data (Open-Meteo)

Hourly weather observations were obtained from the Open-Meteo Historical Weather API [Open-Meteo, 2024] for each sensor location, covering temperature, precipitation, wind speed, cloud cover, and relative humidity.

## 2.2 Preprocessing Pipeline

how did we process NA and extremes

### 2.2.1 Temporal Alignment

All data sources are aligned to the period August 2019 to December 2024. The lower bound corresponds to the earliest available fietstellingen count data in this study, while the upper bound is determined by the coverage of the Statbel accident dataset, which extends to December 2024.

confirm with team

### 2.2.2 Geographic Filtering and Category Refinement

Accident records are filtered to retain only those occurring within the Flemish Region and involving at least one cyclist as a road user. This ensures that the analysis is scoped to the geographic area covered by the AWV sensor network and focuses exclusively on cycling-related safety risks.

### 2.2.3 Spatial Buffer Join

Accident coordinates, originally provided in the Belgian Lambert projection (EPSG:31370), are converted to WGS84 (EPSG:4326) prior to spatial joining. Accident records are then matched to sensors using a BallTree radius search within a 1km threshold . A threshold of 1km was selected over the stricter 500m radius, which yielded only 908 matched accidents with 109 of 151 sensors having fewer than 10 matched crashes — insufficient signal for monthly modelling. At 1km, 2,734 accidents are matched across 148 sensors, with only 53 sensors having fewer than 10 matched crashes.

double-counting as limitation?

# 3 Exploratory Data Analysis

# 4 Methodology

## 4.1 Data Aggregation and Monthly Dataset Construction

## 4.2 Target Variable Definition

tbc

The target variable is a binary indicator of high cycling risk at each sensor location. Risk rate is first computed as:

$$\text{risk\_rate}_i = \frac{\text{accident\_count}_i}{\text{total\_traffic}_i} \tag{1}$$

where $\text{accident\_count}_i$ denotes the number of bicycle accidents recorded within 500 metres of sensor $i$ during the target period (2023–2024), and $\text{total\_traffic}_i$ denotes the total cyclist count recorded by sensor $i$ during the historical period (2019–2022) as a proxy for local exposure. This normalisation aims to identify locations where accident frequency is disproportionate to cycling activity.

To mitigate instability in risk estimates for sensors with low traffic volume, Bayesian smoothing is applied, shrinking individual estimates towards the global mean:

$$\text{risk\_rate\_smoothed}_i = \frac{\text{accident\_count}_i + \alpha \cdot \mu}{\text{total\_traffic}_i + \alpha} \tag{2}$$

where $\mu$ is the global mean risk rate and $\alpha$ is a smoothing parameter. A sensor is then classified as *high risk* if its smoothed risk rate exceeds the global mean:

$$\text{high\_risk}_i = \mathbf{1}\left[\text{risk\_rate\_smoothed}_i > \mu\right] \tag{3}$$

This formulation yields a binary classification task, where the model predicts whether a given sensor location is structurally high risk based on historical traffic and weather features.

## 4.3 Feature Engineering and Lag Structure

## 4.4 Leakage Removal

## 4.5 Model Selection and Training

## 4.6 Threshold Tuning

## 4.7 Evaluation Metrics

# 5 Results

## 5.1 Model Comparison

## 5.2 Threshold Tuning Results

## 5.3 Feature Importance Analysis

## 5.4 Historical Safety Rankings

## 5.5 Forecast Safety Rankings

# 6 Web Application

## 6.1 Interactive Risk Map

## 6.2 Historical and Forecast Safety Rankings

## 6.3 Station-Level Detail Panel

## 6.4 Prediction Engine

# 7 Discussion

## 7.1 Interpretation of Results

## 7.2 Radius Selection

## 7.3 Monthly versus Quarterly Forecasting

## 7.4 Limitations

# 8 Conclusion and Future Work

# References

Agentschap Wegen en Verkeer. Gevaarlijke punten, 2025. URL https://wegenenverkeer.be/veilig-op-weg/gevaarlijke-punten. Accessed: 2026.

Open-Meteo. Open-meteo weather API, 2024. URL https://open-meteo.com/. Accessed: 2026.

Statbel. Geolocation of road traffic accidents 2017–2024, 2025. URL https://statbel.fgov.be/en/open-data/geolocation-road-traffic-accidents-2017-2024. Accessed: 2026.

Statistiek Vlaanderen. Road casualties in the flemish region, 2024. URL https://www.vlaanderen.be/en/statistics-flanders/mobility/road-casualties. Accessed: 2026.

# A GitHub Repository

# B Environmental Setup and Key Dependencies

# C Full Feature List

# D Additional Figures